MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

QP CODE:24900041 - Multi-Disciplinary Course- **MG1MDCPES101 -PHYSICAL FITNESS AND HEALTHY LIVING**

Duration: 1 Hours Maximum Marks: 35

Part A Multiple Choice Questions (1Marks)

1.Which is a type of health-related physical fitness? (K) (1)

Ans: (B) Cardiovascular Endurance

2. What type of exercise would you suggest for improving agility? (A) (1,2)

Ans: (B) Quick Lateral Movements and Cone Drills

3. Which of the following is a test of balance? (K) (3)

Ans: (C) Single-Leg Stand Test

4. Cosmetic fitness primarily focuses on (U) (2)

Ans: (B) Improving Body Aesthetics and Appearance

5. Choose how high-altitude training improve an athlete's performance at sea level? (U) (5)

Ans: (A) By increasing the efficiency of Oxygen Utilization

6.Name the muscle which is responsible for the movement of the arm at the shoulder? (U) (4)

Ans: (D) Deltoid

7. Which type of training is most effective for improving neuromuscular coordination? (U) (5)

Ans: (A) Resistance Training

8. What does the acronym FIIT stand for in fitness training? (K) (3)

Ans (A) Frequency, Intensity, Time and Type

9. Which of the following is a sign of Overtraining? (A) (4)

Ans (C) Fatigue and Decreased Performance

10. How can a fitness enthusiast best apply the principles of recovery in their training regimen? (A) (4)

Ans: (B) Include rest days and lighter workout sessions **(10x1=10)**

Part B Short Answer Type Questions (2 Marks)

11. Design a training program that focuses on improving agility for soccer players. Include specific drills and exercises (A) (3)

We can improve the agility of soccer players through explosive movement, quick directional changes and targeting key components like quickness, balance, coordination, reaction time, and strength. Specific drills include Ladder Drills, Cone Drills, Sprint-Change-Sprint Drills and Reaction-Based Drills

12.What role does the Cardiovascular exercises play in a cosmetic fitness regimen? (A) (2)

Cardiovascular exercises contribute to both the aesthetic and functional aspects of physical appearance. It reduces body fat, improve body composition, increased muscle definition, skin health, posture and confidence.

13. What role doses exercise play in reducing the risk of diseases such as Alzheimer's and Parkinson's (U) (5)

Exercise increases cerebral blood flow, ensuring that the brain receives an adequate supply of oxygen and nutrients. This helps protect brain cells from damage and promotes repair mechanisms. Exercise has anti-inflammatory effects, lowering levels of pro-inflammatory cytokines that can harm brain cells over time. Exercises lower cortisol levels and reduce symptoms of anxiety and depression, which are often precursors or co-morbidities of neurodegenerative diseases.

14.What are the common symptoms of digestive disorders? (K) (4)

Digestive disorders encompass a wide range of conditions affecting the gastrointestinal (GI) tract, and their symptoms can vary depending on the specific condition. Some common symptoms are abdominal pain and discomfort, fatigue, diarrhea, constipation, loss of appetite, bloating and gas, nausea and vomiting, color and texture change of stool.

15.What is the principle of optimum load in the context of exercise? (U) (3)

The principle of optimum load in exercise refers to applying the right amount of stress (load) or resistance to the body during exercise to stimulate desired adaptations without causing injury, overtraining, or burnout. Key elements of optimum load are individualization, progressive overload, avoid over training and keep a balance between load and recovery.

16.List three risk factors associated with Physical inactivity? (An) (2)

Physical inactivity contributes to conditions like high blood pressure, type II diabetes, osteoporosis, arthritis, chronic lower back pain, elevated cholesterol levels and obesity.

17.How does rest contributes to muscle growth and repair? (A) (4)

Rest is a critical component of muscle growth and repair, as it allows the body to recover from the stress of exercise and rebuild stronger. It allows our body to heal and rebuild muscle (Hypertrophy) tissue that's damaged during exercise**.** **(5x2=10)**

Part C Short Essay Type Questions (5Marks)

18. Discuss the psychological benefits of engaging in regular physical activity? (U) (1)

Engaging in regular physical activity has profound psychological benefits that enhance mental well-being, emotional resilience, and cognitive function. Improved mood, release of endorphins (happy hormone) and reduction of stress hormones (cortisol and adrenaline), antidepressant effect, distracts from worries, better stress management, improved memory, increased self-esteem and self-confidence.

19.Define Skill –related physical fitness and explain its significance in athletic performance? (K) (3)

Skill-related fitness refers to abilities that help people learn skills or improve sports performance. It has six components agility, balance, coordination, power, reaction time, and speed. Skill-related fitness enhances sport-specific abilities by making athletes more effective in competition, reduces the risk of injury by improving control, balance, and coordination, boosts confidence as athletes achieves greater proficiency in their movements and provides a competitive edge.

20.What are the common respiratory diseases? Recommend the different activities for controlling the risk factors of respiratory diseases? (K) (5)

Respiratory diseases affect the airways, lungs, and other structures involved in breathing. They can range from mild to severe and may be acute or chronic. Some of the most common respiratory diseases are asthma, Chronic Obstructive Pulmonary Disease (COPD), Pneumonia, Tuberculosis (TB), Bronchitis and Influenza (Flu). Controlling risk factors for respiratory diseases involves engaging in activities that promote lung health, enhance respiratory function, and reduce exposure to harmful substances. Here are some recommended activities to help minimize these risks. Regular Physical Activity, practice breathing exercises, Avoid Smoking, reduce environmental Pollutants, allergens, and irritants, Strengthen the Immune System, Practice Good Hygiene, Maintain a Healthy Weight, Stress Management, Regular Health Check-Ups.

21.How do rest days contribute to the principles of fitness, particularly in terms of recovery and performance enhancement? (A) (4)

Rest days are an essential component of any fitness regimen and play a vital role in recovery and performance enhancement. Rest days allow the body to repair and rebuild, addressing the wear and tear caused by exercise. This aligns with the principle of recovery, which emphasizes that sufficient time is needed for muscles, tendons, and ligaments to heal and strengthen. ***Energy Replenishment****:* Glycogen stores, the body's primary energy source during exercise, are restored during rest periods, ensuring energy availability for subsequent workouts.

Injury Prevention: Overtraining without adequate rest can lead to fatigue, which increases the risk of injuries such as strains, sprains, and stress fractures.

22. Explain the role of exercise in mitigating the effects of the ageing process. How do different types of exercises (e.g., Strength Training, Aerobics) benefit older adults? (Ap) (5)

Regular exercise is one of the most effective ways to combat the effects of aging, improving both physical and mental well-being. Engaging in Strength training can enhance strength and potentially reverse or delay the decline in muscle mass (Sarcopenia) and strength that typically occurs with aging. Additionally, aerobic exercise can improve endurance. Other benefits of exercises to older adults are maintaining independence, reducing the risk of disease, improving mental health, improving cognitive function, strengthening social ties, Reducing the risk of falls

(3X5=15)

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